

JOHNSTOWN CASTLE, Ireland

WMO: 039560

Lat: 52.2978N

Lon: 6.4967W

Elev: 174

StdP: 14.60

Time Zone: 1.00 (IRL)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.7	32.5	24.6	18.8	33.3	26.9	21.0	35.1	29.5	46.4	26.5	45.8	8.1	340	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	69.1	61.6	66.7	60.5	65.0	59.6	63.2	66.5	62.2	65.0	61.1	63.7	9.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.1	84.0	64.2	61.0	80.8	63.1	60.0	77.8	62.3	28.8	66.4	28.0	65.2	27.3	63.8	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.3	21.3	19.0	DB	27.8	73.2	2.6	1.7	25.9	74.4	24.4	75.4	23.0	76.4	21.1	77.6
			WB	26.3	65.6	2.6	1.2	24.4	66.5	22.9	67.2	21.4	67.9	19.5	68.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	43.0	42.7	43.9	47.3	52.0	56.6	59.8	59.0	56.3	52.3	46.5	43.9	(6)
	DBStd	7.35	4.18	4.25	4.05	3.68	3.80	3.55	3.20	2.60	3.12	3.97	4.85	5.15	(7)
	HDD50	1072	217	207	192	95	20	1	0	0	1	23	122	194	(8)
	HDD65	5362	682	626	654	531	402	252	165	187	262	394	554	653	(9)
	CDD50	1188	1	1	3	13	83	200	303	278	189	94	18	5	(10)
	CDD65	3	0	0	0	0	0	1	2	0	0	0	0	0	(11)
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14) Wind	WSAvg	9.5	10.9	10.4	9.9	9.3	9.3	8.5	8.4	8.8	8.6	9.3	9.8	10.9	(14)
(15) Precipitation	PrecAvg	38.5	3.9	3.0	2.5	2.5	2.6	2.6	2.7	3.0	3.0	4.4	4.2	4.1	(15)
	PrecMax	52.5	8.8	7.4	5.0	4.9	6.7	7.6	6.2	10.5	6.2	9.6	9.1	10.9	(16)
	PrecMin	29.5	0.9	0.7	0.6	0.7	0.2	0.4	0.5	0.2	0.6	1.0	1.2	1.2	(17)
	PrecStd	5.5	1.7	1.6	1.3	1.3	1.4	1.7	1.5	2.2	1.5	2.0	2.1	2.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	53.5	56.8	61.1	66.7	71.4	72.8	69.5	66.7	62.1	57.5	54.6	(19)
		MCWB	51.2	50.3	50.1	53.8	59.2	61.5	63.5	62.1	60.3	58.7	56.0	53.6	(20)
	2%	DB	51.8	51.4	53.5	57.4	62.9	67.3	69.8	66.5	64.5	60.5	56.4	53.5	(21)
		MCWB	50.7	49.2	48.8	51.4	57.2	59.9	62.2	60.7	59.1	58.0	55.0	52.4	(22)
	5%	DB	50.8	50.3	51.4	55.2	60.3	64.7	67.7	65.0	62.9	59.4	55.3	52.7	(23)
		MCWB	49.6	48.7	48.1	50.3	55.3	58.6	61.2	59.8	58.5	57.2	53.8	51.5	(24)
	10%	DB	49.7	49.4	50.0	53.0	58.3	62.8	65.6	63.8	61.6	58.2	54.1	51.5	(25)
		MCWB	48.4	47.8	47.4	48.9	53.6	57.4	60.2	59.2	58.0	56.3	52.5	50.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	51.5	52.0	54.9	60.3	63.1	65.5	63.9	63.0	60.3	56.8	54.3	(27)
		MCDB	52.3	52.4	54.0	60.1	65.5	68.9	69.9	66.2	64.6	61.1	57.1	54.4	(28)
	2%	WB	51.2	50.3	50.5	52.7	58.0	61.1	63.8	62.7	61.2	59.2	55.4	53.0	(29)
		MCDB	51.5	51.0	51.9	56.2	61.8	65.3	67.7	64.6	62.6	60.0	55.9	53.4	(30)
	5%	WB	50.1	49.3	49.2	51.2	56.3	59.7	62.6	61.8	60.1	58.0	54.1	51.9	(31)
		MCDB	50.6	50.1	50.7	53.8	59.5	63.3	66.1	63.6	61.6	58.9	55.0	52.5	(32)
	10%	WB	48.8	48.2	48.0	50.1	54.8	58.5	61.4	60.8	59.1	56.9	52.7	50.7	(33)
		MCDB	49.7	49.3	49.6	52.3	57.6	61.8	64.6	62.7	60.8	57.9	53.9	51.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.5	9.9	10.3	10.3	10.4	10.3	10.0	9.9	9.0	8.5	8.1	(35)
		MCDBR	8.3	8.4	10.9	12.7	12.0	12.6	12.4	10.9	10.9	8.7	8.1	8.2	(36)
	5% WB	MCWBR	8.7	7.3	8.2	8.5	7.8	8.1	7.4	7.0	7.6	7.3	8.1	8.9	(37)
		MCWBR	8.6	7.7	10.1	10.8	11.2	11.4	10.9	9.1	8.7	7.8	7.9	7.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.333	0.344	0.371	0.381	0.389	0.376	0.383	0.363	0.359	0.350	0.327	0.321	(40)
		taud	2.306	2.366	2.317	2.323	2.343	2.409	2.395	2.462	2.469	2.440	2.379	2.284	(41)
	Ebn at Noon		205	243	258	271	274	278	274	274	262	238	208	186	(42)
		Edh at Noon	20	25	32	36	37	35	35	31	28	23	19	18	(43)
(44) All-Sky Solar Radiation	RadAvg		241	463	817	1277	1600	1713	1571	1328	974	554	302	185	(44)
	RadStd		23	35	86	114	102	147	137	85	67	49	29	23	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page